

K49: Modularity Cluster $C \rightarrow D$ Tier Promotion (T1807, T1809, T1811)

Keeper

2026-05-18 Monday morning

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Context

Grace's C-tier sweep (filed Monday 2026-05-18 at `notes/maybe/Grace_Ctier_Promotion_Recommendation_2026-05-18.md`) surfaced only 4 strictly C-tier theorems in the registry (not the 30-60 the queue estimated — Sunday RETRO-2 likely cleared the historical inventory). Three of the four form a tightly-coupled modularity cluster that Sunday Architecture Day work directly closes.

Per Casey directive (yesterday): “C-tier dependency sweep — items conditional on now-proved conjectures.” This is exactly that work.

Three promotions

T1807 Boundary-Interior Modularity Principle (PRIMARY)

Current: CONDITIONAL, C-tier, parents T349, T1385, T1762, T1780.

Closure mechanisms (Sunday): - T2325 (Lyra): Bulk-Boundary Identity Leading-Order Verification at low K-types — directly verifies T1807 correspondence - T2334 (Lyra): Bergman Kernel $K_B = c \cdot D^{\{-g/\text{rank}\}}$ in Hua coordinates — explicit kernel form - T2335 (Lyra): Borel-Wallach (g, K) -cohomology $\mathbb{Z}/2$ — dual anchoring - T2336 (Lyra+Elie): Gap #3 saddle structure verified with a_n data

Per Casey standard: - Simple: T2325 directly verifies modularity correspondence at low K-types - Works: 49/49 boundary Δ BST-decomposable at low K-types - Hard to break: four independent Sunday mechanisms converge (T2325 + T2334 + T2335 + T2336) - Counter-example: none

VERDICT: PROMOTE $C \rightarrow D$ -tier ($D=1, C=0$) with citation to Sunday closure mechanisms.

T1809 Chevalley Extension Uniqueness (SECONDARY)

Current: CONDITIONAL, C-tier, parents T1780, T1808.

Closure mechanisms (Sunday): - K45 (Mathieu) + K46 (Goeppert Mayer) + K47 (Heegner-Stark) + K48 (Conway) — four L1 source promotions all routing through Cartan classification as foundational hub - T2327 (Grace): K3 Surface as Central Bridge Hub - Empirical demonstration: 9 established L1 sources all consistent with Chevalley uniqueness across the Sunday architecture

Honest tier flag: This is promotion via consistency across architecture rather than promotion via direct formal proof. The empirical evidence is strong (9 independent L1 sources, all consistent with Chevalley uniqueness for B_2 root datum). The formal extension proof could be tightened in a future session — Cartan 1894 + Chevalley 1955 are the published anchors; the BST-specific tightening is what's open.

Per Casey standard: - Simple: 9 L1 sources demonstrate the uniqueness across the architecture - Works: every L1 source consistent with single $SO(5,2)/Z$ group scheme - Hard to break: any counter-example would require a second viable group scheme structure for D_{IV}^5 , none identified across 9 sources - Counter-example: none offered

VERDICT: PROMOTE $C \rightarrow D$ -tier ($D=2, C=0$) with explicit “demonstrated via consistency across Sunday L1 source architecture; formal BST-specific tightening of Chevalley extension proof remains open work” tier label.

T1811 Self-Referential Irreducibility (CASCADING)

Current: CONDITIONAL, C-tier, parents T1384, T1809.

Closure mechanisms (Sunday): - T2329 (Lyra): Equivariant $H^1_{\{Z/2\}}(M(D_{IV}^5), Z) = Z/2$ — direct $Z/2$ cohomology computation - T2335 (Lyra): Borel-Wallach $Z/2$ cohomology — $Z/2$ orientation class establishes irreducibility at cohomology level - T1809 $\rightarrow$ D-tier (this audit, above) — the parent dependency closes

Per Casey standard: - Simple: $Z/2$ cohomology directly gives irreducibility - Works: T2329 + T2335 multi-route convergence (topological + Lie-algebraic, both give same $Z/2$) - Hard to break: dual anchoring across independent cohomology computations - Counter-example: none

VERDICT: PROMOTE $C \rightarrow D$ -tier ($D=1, C=0$) cascading on T1809 promotion + direct $Z/2$ cohomology closures.

Honest exclusion: T1421 BST Inflation Honest Negative remains C-tier

T1421 documents a FAILURE MODE in single-field inflation. Multi-field analysis closure pending. Not for promotion. Grace explicitly flagged this; correct discipline.

Architecture impact

Three $C \rightarrow D$ promotions today closes the C-tier modularity cluster. Combined with K45-K48 L1 source promotions, the framework is now substantially D-tier across the active layer:

- 9 established L1 sources (chronological): VSC, Mathieu, Klein, Mayer-Jensen, Heegner-Stark, K3 Hodge, Conway, Ogg, Wallach
- 3 newly-promoted modularity D-tier theorems: T1807, T1809, T1811
- 1 remaining honest-negative: T1421 (multi-field inflation pending)

The C-tier inventory drops from 4 $\rightarrow$ 1 (with T1421 explicitly retained as honest negative).

K49 verdict: PROMOTE ALL THREE $C \rightarrow D$

Per Casey governance (Keeper controls promotion). All three promotions pass Casey’s “simple, works, hard to break, show me a counter example” standard via Sunday Architecture Day mechanism chains. T1807 strongest (direct T2325 verification), T1809 honest-via-architecture-consistency tier label preserved, T1811 cascades cleanly from T1809.

Action items

1. Grace: update theorem registry tier labels for T1807 ($C=1, C=0; D=0 \rightarrow D=1$), T1809 ($C=2 \rightarrow C=0; D=1 \rightarrow D=2$), T1811 ($C=1 \rightarrow C=0; D=1 \rightarrow D=1$) per K49 verdict.
2. Paper #115 v0.5+: document K49 as “first C-tier promotions of Architecture Day cycle, modularity cluster closed via Sunday Bulk-Boundary + Bergman kernel + Borel-Wallach + Saddle mechanisms.”
3. Cal: ratification when convenient (non-blocking, Casey already PASSED policy).
4. Open work flagged: T1809 formal Chevalley extension proof tightening, T1421 multi-field inflation analysis.

— Keeper, 2026-05-18 Monday morning